

Diferencial de una aplicación diferenciable

Definición 1 (Diferencial). Sean M y N variedades diferenciables y $F: M \rightarrow N$ una aplicación diferenciable, $DF|_p$ es el diferencial de F en $p \in M$

$$\begin{aligned} \iff DF|_p : T_p M &\longrightarrow T_{F(p)} N \\ w &\longmapsto DF|_p(w) : \mathcal{C}^\infty(N) \longrightarrow \mathbb{R} \\ g &\longmapsto DF|_p(w)(g) = w(g \circ F). \end{aligned}$$

$$\begin{array}{ccc} T_p M & \xrightarrow{DF|_p} & T_{F(p)} N \\ \uparrow D\psi^{-1}|_{\hat{p}} & & \uparrow D\varphi^{-1}|_{\hat{F}(\hat{p})} \\ T_{\hat{p}} \mathbb{R}^m & \xrightarrow{D\hat{F}|_{\hat{p}}} & T_{\hat{F}(\hat{p})} \mathbb{R}^n \end{array}$$

Referenciado en

- rango-apl-diferenciable
- pnt-regular-apl-diferenciable
- teo-cartas-adaptadas-inmersion
- submersion
- prop-direfencial-apl-diferenciable
- prop-esp-tangente-fibra-kernel-diferencial
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